

NAME: _____ PDG: _____ Register No: _____



ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2011

CHEMISTRY

8872/02

Higher 1

19 September 2011

Paper 2 Structured Questions

2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet
 Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate writing paper.

Start each question on a fresh sheet of paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use						
Paper 1 (33%)	Paper 2 (67%)					
	Section A			Section B		Total
	Q1	Q2	Q3			
/ 30						/ 80
				Final marks		/ 100
				Grade		

This document consists of **14** printed pages.

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 (a) The element nitrogen has two naturally occurring isotopes, ^{14}N and ^{15}N . Their natural abundance is 99.63% and 0.37% respectively.

(i) Define the term *relative isotopic mass*.

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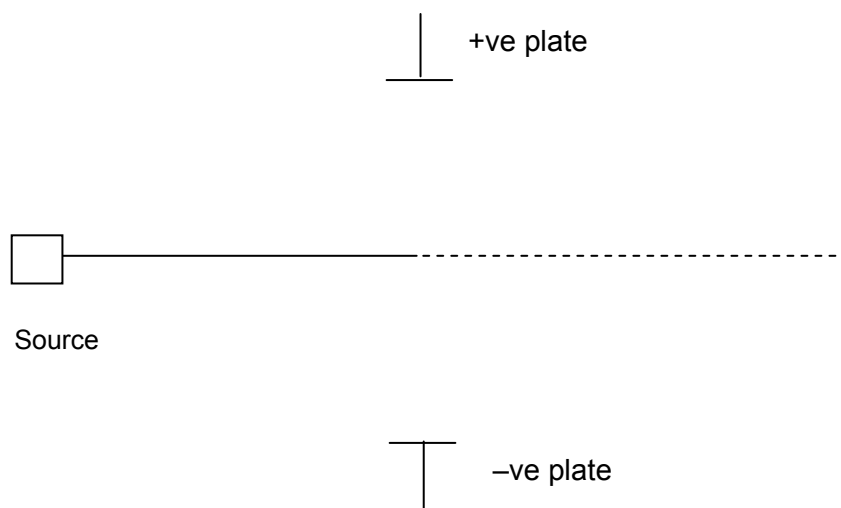
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(ii) Calculate the relative atomic mass for nitrogen, giving your answer to 5 significant figures.

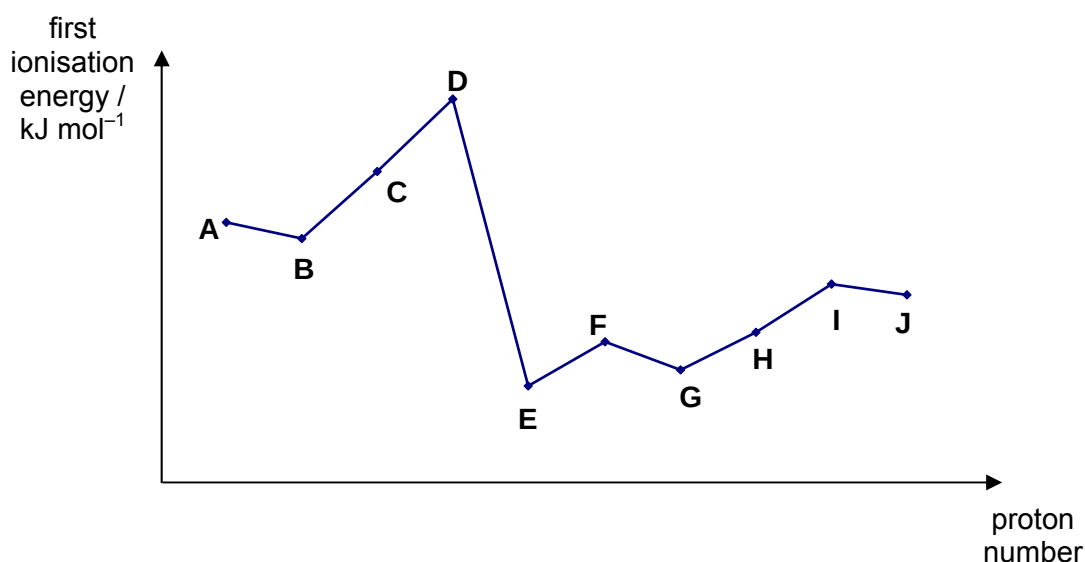
- (iii) A gaseous mixture of $^{14}\text{N}^{3-}$ and $^{15}\text{N}^{3-}$ is passed through an electric field which is at right angles to their direction of travel. On the diagram below, show and label clearly how the beams of the two particles are deflected when subjected to an electric field, and explain the phenomenon.

You should relate clearly the *magnitude* and the *direction* of deflection of each beam to the other.



[4]

- (b) Nitrogen is a second period element. The first ionisation energies of ten successive elements **A** to **J** in both Periods 2 and 3 of the Periodic Table are shown below:



- (i) Which one of the ten elements represents nitrogen?

- (ii) Element **B** and **J** are in the same group.

Using the electronic configurations of **B** and **J**, explain why the first ionisation energy of **J** is found to be lower than that of **B**.

B: **J:**

- (iii) Identify the Group of the Periodic Table to which element **F** belongs. Using the symbol **F**, suggest the formula of a compound that element **F** will form with hydrogen.

Group:

Formula of compound:

- (iv) Element **I** forms more than one stable chloride. State the formula of **one** of the chlorides and predict its shape.

Draw dot-and-cross diagram to show the arrangement of electrons in the chloride of **I** that you have stated using the symbol **I**.

Formula of chloride:	Shape:
Dot-and-cross diagram:	

- (v) In general, ionisation energy increases across a period. However, there are some discontinuities in the graph shown above. One of the exceptions in the general trend occurs between elements **I** and **J**.

Explain why the first ionisation energy of **I** is higher than that of **J**.

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[9]

- (c) Nitrogen is used in the Haber Process to form ammonia, NH_3 , which is an important ingredient in the manufacture of fertilisers. NH_3 has similar molecular mass compared to methane, CH_4 , but both compounds have very different solubilities in water.

Describe the bonding present in each compound and explain the variation in the solubility of these two compounds in water.

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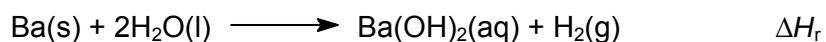
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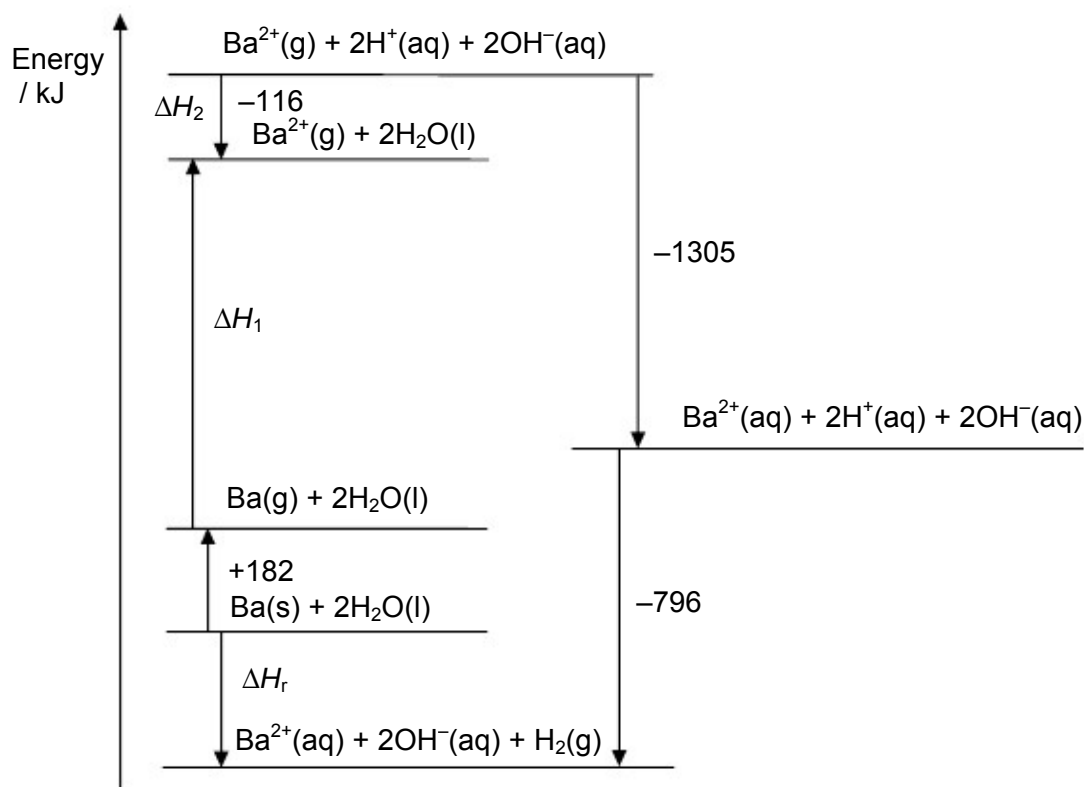
[2]

[Total: 15]

- 2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is used as an additive in thermoplastics and PVC stabilisers to improve plastic properties. This material is also used as a general purpose additive for lubricants and greases. It is formed by the reaction of barium with water, as shown in the following reaction.



The energy cycle for the formation of $\text{Ba}(\text{OH})_2$ is given below.



- (a) (i) What do ΔH_1 and ΔH_2 represent?

ΔH_1

ΔH_2

- (ii) Using the *Data Booklet*, calculate a value for ΔH_1 .

- (iii) Hence, using the energy cycle, calculate the enthalpy change of reaction of barium with water, ΔH_r .

- (iv) In an industrial laboratory, a researcher performed the experiment and added barium to 100 cm³ of water. The temperature of the resulting solution rises from 22.0 °C to 33.5 °C. Using your answer in (iii), calculate the mass of barium added.

Assuming that the specific heat capacity of the solution is 4.18 J g⁻¹ K⁻¹ and that the density of the solution is 1.00 g cm⁻³.

- (v) Draw an energy profile of the reaction of barium with water, showing clearly the activation energy and ΔH_r .

- (b) (i) Define standard enthalpy change of formation, ΔH_f^θ .

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- (ii) Given that enthalpy change of formation of water, $\Delta H_f^\theta \text{ H}_2\text{O(l)}$ is $-285.8 \text{ kJ mol}^{-1}$, and using your answer in (a)(iii), calculate the enthalpy change of formation of $\text{Ba(OH)}_2\text{(aq)}$.

[3]

- (c) (i) Write an equation, including state symbols, for the lattice energy of solid Ba(OH)_2 .

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- (ii) State, with reason, if the lattice energy of Ba(OH)_2 is more or less exothermic than Sr(OH)_2 .

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[2]

[Total: 12]

- 3 Organic solvents are among the most useful industrial chemicals. Dichloromethane, CH_2Cl_2 and ethane-1,2-diol, $\text{HOCH}_2\text{CH}_2\text{OH}$ are common solvents used in the paint industry.

(a) For each of these two solvents, describe a test to show the presence of each compound type. You should also include the reagents and conditions necessary for each reaction. State the expected observations for a positive test.

Compound	Description of test	Observations
CH_2Cl_2		
$\text{HOCH}_2\text{CH}_2\text{OH}$		

[4]

- (b) Under suitable conditions, dichloromethane can be hydrolysed to produce a toxic gas with empirical formula CH_2O .

Compounds with the empirical formula CH_2O can have the molecular formula $\text{C}_2\text{H}_4\text{O}_2$.

In the space below, draw the *displayed* formula of a structural isomer of $\text{C}_2\text{H}_4\text{O}_2$ which will exhibit the respective property.

None of the isomers contain rings or O–O bonds.

- (i) Isomer **K** reacts with Na_2CO_3 .

(ii) Isomer **L** reacts with aqueous bromine.

(iii) Isomer **M** reacts with Fehling's solution.

(iv) Isomer **N** has no reaction with the reagents in (i), (ii) and (iii).

[4]

(c) Compound **P**, containing three carbons, also has an empirical formula CH_2O .

P gives an effervescence when added to NaHCO_3 . When heated with acidified KMnO_4 , **P** forms compound **Q**. **Q** gives an orange precipitate with 2,4–dinitrophenylhydrazine. Both **P** and **Q** give a pale yellow precipitate, **R**, when warmed with alkaline aqueous iodine separately.

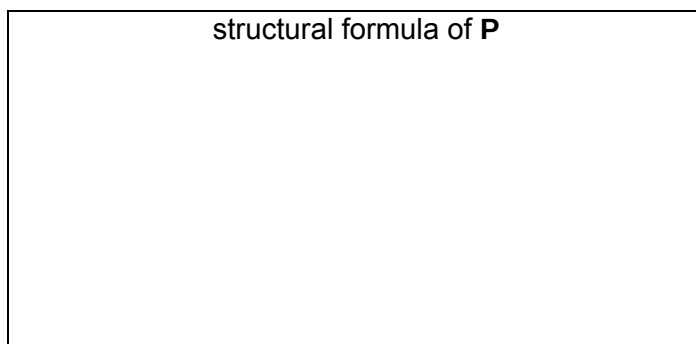
(i) What is the identity of **R**?

Identity of **R**:

(ii) State the functional group(s) present in **Q**.

Functional group(s):

(iii) Draw the structural formula of **P**.



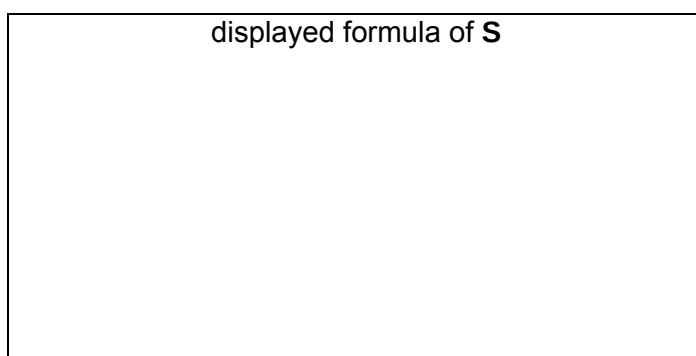
[3]

(d) When warmed with concentrated H_2SO_4 , compound **P** forms a cyclic compound **S**. Compound **S** has a molecular formula $\text{C}_6\text{H}_8\text{O}_4$ and it shows no reaction with the reagents stated in (c).

(i) State the type of reaction that has occurred in the conversion of **P** to **S**.

Type of reaction:

(ii) Suggest a displayed formula for **S**.



[2]

[Total:13]

Section B

Answer **two** questions from this section on separate writing paper.

- 4 During exercise, muscle cells convert glucose into lactic acid. The lactic acid is used in the *mitochondria*, the energy factories in muscle cells, to produce energy. To help athletes optimize their training regime, sports scientists analyse the lactic acid content in their blood during training to determine how effective their muscle *mitochondria* takes up the lactic acid produced. A sample of blood is taken from the athlete and the monoprotic acid is extracted and subsequently titrated with a base.

(a) Outline the *Brønsted–Lowry* theory of acids and bases.

[2]

(b) A solution of lactic acid with concentration 0.10 mol dm^{-3} has a pH of 2.43.

(i) Is lactic acid a strong or weak acid? Explain your answer.

(ii) Use the data given to calculate the value of K_a for lactic acid.

(iii) Suggest a suitable indicator for the titration of lactic acid with aqueous calcium hydroxide.

(iv) A sample of lactic acid having a mass of 1.00 g was extracted from a blood sample. It was dissolved in water and titrated with $0.250 \text{ mol dm}^{-3}$ calcium hydroxide. It was found that 22.3 cm^3 of hydroxide was required for neutralisation.

Calculate the M_r of lactic acid.

(v) The mass composition of carbon, hydrogen and oxygen in lactic acid is 40.0%, 6.7% and 53.3% respectively. Determine the empirical formula of lactic acid.

Hence deduce the molecular formula of lactic acid using your answer in (iv).

(vi) When lactic acid reacts with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, it gives an organic product with the same number of oxygen atoms. Draw the displayed structural formula of lactic acid.

[11]

(c) State what is meant by a *buffer solution*.

[2]

(d) Explain, using equations why an aqueous mixture of lactic acid and sodium lactate can act as a buffer solution

(i) on addition of acid,

(ii) on addition of alkali.

You may represent lactic acid as HA and sodium lactate as Na^+A^- .

[2]

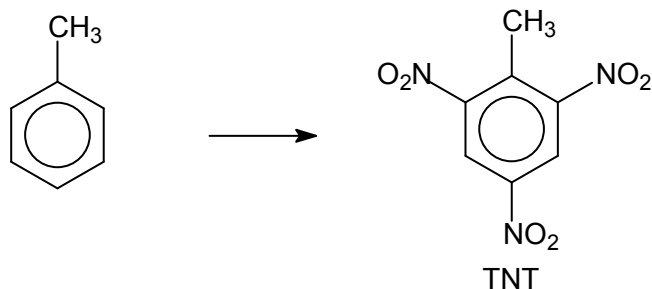
(e) A buffer solution best resists pH changes when the concentrations of lactic acid and that of sodium lactate are the same

Determine the mass of sodium lactate that needs to be added to 500 cm^3 of 0.1 mol dm^{-3} solution of lactic acid to create this buffer solution.

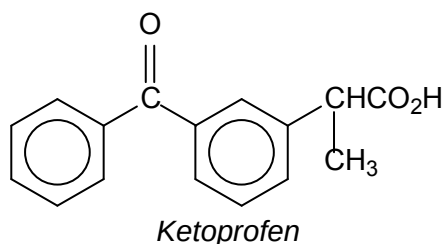
[3]

[Total: 20]

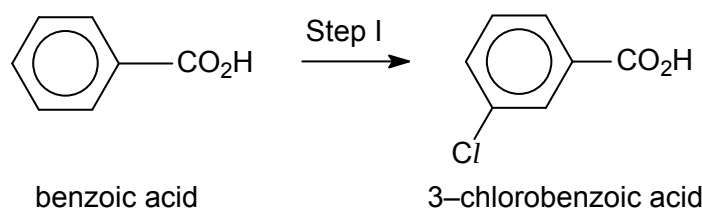
- 5 Methylbenzene is an important raw material for the manufacture of *trinitrotoluene*, which is commonly known as TNT. TNT is one of the most used explosives for military and industrial applications. It is valued partly because of its insensitivity to shock and friction, which reduces the risk of accidental detonation.



- (a) The synthesis of TNT consists of a series of reactions involving methylbenzene.
- What is the type of reaction involved when methylbenzene is converted to TNT?
 - One advantage of TNT is that it can be used in wet environments because it does not dissolve in water. Suggest a reason for its solubility in water.
- [2]
- (b) Methylbenzene can be used to make *Ketoprofen*, a substitute drug for patients who are allergic to aspirin.

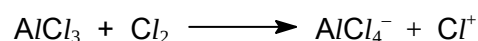


Ketoprofen is synthesized through a series of steps in the Voskian Synthesis. Step I of this synthesis is given below.



- State the reagents and conditions required to convert methylbenzene to benzoic acid.
- State and explain whether methylbenzene or benzoic acid has a higher boiling point.
- Explain why 3-chlorobenzoic acid is a stronger acid than benzoic acid.

In the Step I given above, chlorine is added to benzoic acid in the presence of anhydrous aluminium chloride. The reaction between chlorine and aluminium chloride is shown below.



- Suggest why the reaction between AlCl_3 and Cl_2 aids in the rate of formation of 3-chlorobenzoic acid.

[6]

- (c) In order to determine the rate equation for the Step I of Voskanian Synthesis, an investigation using the same concentration of chlorine was first carried out at constant temperature to find the relationship between initial rate and initial concentration of benzoic acid.

- (i) Explain what is meant by *order of reaction*.
- (ii) Experimental data showed that the first step of the Voskanian Synthesis was first order with respect to benzoic acid.

Sketch a graph of initial rate against initial concentration of benzoic acid.

- (iii) The concentration of chlorine was halved and a new experiment was carried out at the same temperature. When a similar graph was obtained, the gradient at each point was half that obtained from (ii).

What is the order with respect to Cl_2 ? Explain your answer.

- (iv) Hence suggest a rate equation for the first step of the Voskanian Synthesis.
- (v) What are the units of the rate constant for the rate equation you have suggested in (iv)?
[7]

- (d) Explain, with the aid of a sketch of the Boltzmann distribution, how the presence of a catalyst can cause an increase in the rate of a chemical reaction.

[5]

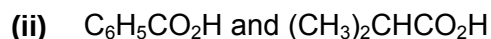
[Total: 20]

- 6 (a) Explain why ethanoic acid is a stronger acid than ethanol.

[2]

- (b) Acids are one of the products that could be formed from the oxidative cleavage of alkenes.

Suggest the structure of each alkene that gives the following acid(s) as the only carbon-containing product upon undergoing oxidative cleavage.



[2]

- (c) Fruits usually contain a mixture of organic acids. Compound **T**, $\text{C}_5\text{H}_8\text{O}_5$, is an acid found in blueberries. 0.37 g of **T** required 25.0 cm^3 of 0.20 mol dm^{-3} aqueous NaOH for complete neutralisation. **T** shows no reaction with either hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or with aqueous bromine. 1 mol of **T** reacts completely with 3 mol of sodium metal.

Use all of the information provided to identify compound **T** and draw its structural formula. Clearly show all the deductions that you make.

[7]

- (d) Compound **U** is an alcohol. A colourless, neutral liquid **V** ($M_r = 232$) is formed when 1 mol of **T** reacts completely with 2 mol of **U**.

Heating **U** with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ gives **W**, a compound often found in ripe fruits.

An orange–yellow precipitate is formed when 2,4–dinitrophenylhydrazine is added to **W**. **W** has no reaction with Tollen's reagent.

- (i) Suggest suitable reagents and conditions for the reaction between **T** and **U**.

- (ii) Deduce the M_r of **U**. Show your reasoning clearly.

- (iii) Hence identify compounds **U**, **V** and **W** from all the information provided. Explain your reasoning.

[9]

[Total: 20]